

Names: _____

Section: _____

CS471 – Final Project Scoping Worksheet

Artificial Intelligence: Final Project Design

Project Timeline & Expectations (250 Points Total)

This project requires you to build, test, and analyze an AI agent solving a tactical military problem. The milestones are designed to prevent feature creep and ensure you focus on the underlying algorithmic logic.

- **Lesson 32: Project Design (25 pts):** *The Scope Contract.* You will complete and submit this worksheet. The goal is to define a strict Minimum Viable Product (MVP) utilizing exactly one core CS471 algorithm. Your instructor will review this to ensure the project is mathematically feasible and appropriately scoped before you begin coding.
- **Lesson 36: Project Checkpoint (50 pts):** *The "Ugly but Functional" Milestone.* You must demonstrate the core mathematical algorithm executing without crashing. A Graphical User Interface (GUI) is completely optional, and terminal output is acceptable. The priority is functional algorithmic logic (e.g., the A* loop completes, the Q-table updates).
- **Lesson 41: Final Project & Presentation (175 pts):** *The Defense.* (150 pts for the final deliverable, 25 pts for the presentation). You will demonstrate the final AI agent and present your operational analysis. You must prove your AI learned or optimized its task using mathematical metrics (e.g., reward convergence graphs, confusion matrices, Precision-Recall tradeoffs) and articulate the tactical implications of its successes and failures.

Teams

This is a group with teams of at least **two** and no more than **three** members. Each member will be evaluated based on their contribution by both the members of their team and by the instructor. If you do not contribute to the project, your grade will be negatively impacted.

Ideation Matrix

If you do not have a project idea, select an archetype from the menu below based on the algorithms covered through Lesson 37. Then adapt that core topic to the problem you want to solve. You may adapt the specific military scenario to fit your interests.

NOTE: Using AI to assist you in this project is expected; however, you must be able to defend your project and explain what is happening in any code you submit.

CS471 Core Algorithm	Tactical Domain	Example Project Scope (MVP)
A* / Pathfinding	Logistics	Optimal supply convoy routing across dynamic, 2D tile terrain with changing movement cost heuristics (e.g., mud, threats).
Adversarial Search	Wargaming	A Minimax agent with alpha-beta pruning playing a simplified zero-sum board game of network node capture.
Reinforcement Learning	Auto-Flight	A Q-learning agent discovering optimal flight paths through a stochastic environment (e.g., moving radar threat zones).
Bayesian Networks	Threat Assessment	A probabilistic diagnostic engine taking in noisy sensor evidence to calculate the probability of an adversary's hidden intent.
Hidden Markov Models	SIGINT	A Viterbi decoder script inferring a submarine's hidden maneuver sequence from a time-series of noisy acoustic pings.
Classical ML (Forest/SVM)	Cyber Defense	A supervised model trained on tabular network traffic to distinguish normal packets from malware, evaluated via PR curves.
Deep Learning (CNNs)	ISR	A Convolutional Neural Network trained to classify overhead satellite imagery, focusing on preventing False Positives.
Autoencoders	Anomaly Detect	An unsupervised network trained on normal base ops data to flag zero-day cyber anomalies via reconstruction error spikes.
Agentic AI / ReAct	Intel Analysis	A text-based reasoning agent that searches a local database of simulated intelligence reports to output a tactical summary.
RAG Pipeline	Knowledge Mgmt	A script that vector-embeds a military doctrine manual, allowing operators to query the text and receive grounded answers.
Explainable AI (XAI)	Verification	Applying SHAP values to a "black box" targeting model to mathematically prove why the AI chose to engage a specific target.

Project Scoping Questionnaire

Complete the following fields to lock in your project scope. Be specific and concise.

1. The Operational Problem: What specific challenge, simulation, or game is your AI agent attempting to solve?

2. The Environment Model: What are the exact states, actions, and observations (or reward structures) your agent will interact with?

3. The Algorithm: Which specific CS471 mathematical framework (e.g., Q-Learning, HMM Filtering, CNN) will drive the agent's decision-making logic?

4. The Baseline (Lesson 36 Checkpoint): What is the simplest, most hardcoded, terminal-only version of this project you can physically demonstrate in four lessons to prove the algorithm works?

5. The Evaluation Metric: How will you mathematically prove your AI learned or optimized its task during the final Lesson 41 presentation (e.g., reward convergence graphs, confusion matrix, total path cost)?